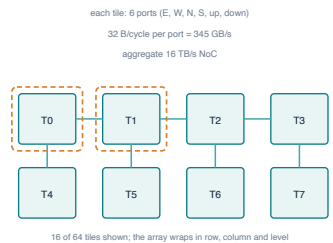
a 3D tile topology



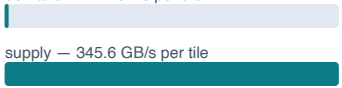
b Halo wait: 46 cycles



Three stencil kernels per step, two exchanges each, fully overlapped with arithmetic on other tiles: 276 cycles hidden in 61,900.

0.5% of the step budget

c Bandwidth: demand vs supply



82× headroom absorbs bursty exchange at kernel boundaries. A shared-L3 implementation would need 4.5× more energy per halo byte and would add 3× the latency.

direct fabric through L3